

Windows Setup Guide

Offline Docker Desktop installation, safe NIC handling, validation, repair, and recovery

v0.3.0 Offline Release | Windows 11 | July 2026

Item	Value
Homeserver URL	Environment-specific (POC: http://10.20.30.2)
Matrix identity	axon.home.arpa
Client networks	Flat LAN or routed/NATed private subnets
Host exposure	TCP 80 only

Axon provides text messaging through stock Element or Element X. Internet, cellular data, federation, media, calling, and internet push notifications are outside the initial proof of concept.

1. Prepare Windows and the LAN

- Windows 11 x64 build 22631 or newer with an administrator account.
- Hardware virtualization enabled in UEFI/BIOS.
- Docker supports an 8 GiB RAM baseline; 16 GiB RAM and 100 GiB free disk are recommended for the planned 200-user deployment. Capacity warnings do not block a normal Axon installation.
- A host with 32 GiB RAM and SSD/NVMe storage provides comfortable headroom.
- Use a connected physical Ethernet or Wi-Fi adapter and extract the complete ZIP to a local NTFS drive.
- Confirm the organization is authorized to accept and use Docker Desktop; some enterprise and government use requires a paid subscription.

Proven routed/NAT example

Device	Example	Role
Upstream gateway	172.20.0.1/24	Upstream LAN gateway
Axon host	10.20.30.2/24	Stable or DHCP-reserved Windows address
Main network gateway	WAN DHCP; LAN 10.30.0.1	Routes/NATs or forwards client traffic
Downstream routers	Static transit addresses	May provide their own client DHCP/NAT

Record the Axon IP, adapter, gateway WAN/LAN addresses, downstream client CIDRs, Element-facing URL, and the source addresses Windows sees. Preserve the correct Windows NIC configuration by default.

2. Verify and install

1. Extract Axon-v0.3.0-offline-win-x64.zip completely.
2. Double-click Install Axon.cmd.
3. Approve the Windows administrator prompt.

- The launcher bypasses PowerShell policy only for its own process. It does not change the machine policy.
- Strict checksums run automatically before bundled installers or images. A failed hash normally means the ZIP must be copied again.
- If Windows features, WSL, or Docker request a restart, restart and double-click Install Axon.cmd again.
- The installer lists connected adapters and IPv4 addresses. The default NicMode Preserve does not alter the selected address, gateway, or DNS.
- Create the first Matrix server administrator when prompted. Use a unique operational password of at least 12 characters.

Routed source scopes

```
.\scripts\Install-Axon.ps1 -BindIp 10.20.30.2 -InterfaceAlias "Ethernet" -AllowedRemoteAddress "10.30.0.0/24", "10.40.0.0/24"
```

Only use NicMode Configure when Axon is explicitly authorized to change the adapter. It adds and verifies the new address before removing older IPv4 addresses.

3. Validate the running stack

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -File .\scripts\Test-Axon.ps1
```

- postgres, synapse, and gateway must all report healthy.
- %ProgramData%\Axon\runtime\synapse\homeserver.yaml must exist and be non-empty.
- http://10.20.30.2/_matrix/client/versions (replace with the selected IP) must return Matrix JSON.
- Ports 8008, 8780, and 5432 must not be reachable through the Axon LAN address.
- Axon Control must answer only at http://127.0.0.1:8780.

4. Cold start and recovery

- After Windows restart, start Docker Desktop and wait for its Linux engine. The axon group normally starts automatically.
- Open the Axon Control Desktop shortcut. If it does not load, run Axon Control Panel from Task Scheduler.
- If GUI login reports Synapse unavailable, start the complete axon group in Docker Desktop, then refresh.
- For an interrupted install, double-click Install Axon.cmd again. Existing runtime secrets and data are preserved.

Repair command

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -File .\scripts\Repair-Axon.ps1
```

Do not delete %ProgramData%\Axon or Docker volumes while diagnosing a failure. Repair preserves secrets and PostgreSQL data.